Appendix 3 solutions

1

Using $\mathbb{N}^+$, let $f(1) = 0, f(2n) = n, f(2n+1) = -n$ for $n \geq 1$, giving $0, 1, -1, 2, -2, \ldots$. Positive k has unique preimage $2k$, negative $-k$ unique preimage $2k+1$, and zero unique preimage one. The index classes are disjoint and cover all positive integers, proving bijectivity.

Alternatively on $\mathbb{N}_0$, use $g(0) = 0, g(2k-1) = k, g(2k) = -k$. Do not mix the conventions.

2

Diagonal zero is $(0, 0)$, indexed zero. Diagonal one is $(0, 1), (1, 0)$, indexed one and two. The next diagonal has indices three, four, five. Diagonal s contains $s+1$ pairs; earlier diagonals contain $1 + \cdots + s = s(s+1)/2$, the pairing offset. Visiting each diagonal in increasing m visits every pair once, giving the countable enumeration. Its unique inverse confirms bijectivity.

3

Choose a distinct listing $S = \{s_1, s_2, \ldots\}$. Finite A is countable by definition. For infinite A, retain exactly its members in the original index order. Infinitely many terms remain, without duplicates, and every element of A occurs. This is a bijective listing by positive natural numbers. The same retention argument for finite S gives a finite list.

4

For two countable nonempty sets, interleave $a_1, b_1, a_2, b_2, \ldots$ and remove duplicates. Empty sets can be skipped. This proves the two-set case; finite induction or finite interleaving proves every finite union. Selecting finitely many enumerations creates no infinite choice issue.

With simultaneously specified $A_n = \{a_{n,1}, a_{n,2}, \ldots\}$, the map $(n, k) \mapsto a_{n,k}$ is onto the union. Its countable pair domain and duplicate removal give a countable union, with empty sets omitted.

If only each set's separate countability is given, ZF need not provide a simultaneous sequence of enumerations. Under standard ZFC or a sufficient Countable Choice assumption, the result follows. Individual countability and a uniform choice of enumerations are distinct information.

Appendix 3 solutions (continued)

5

Use natural numbers starting at zero. Suppose $\mathcal{P}(\mathbb{N}_0) = \{S_0, S_1, \ldots\}$, listing every subset at least once. Put $D = \{k : k \notin S_k\}$. As a subset, it must equal S_m for some m. But $m \in D \iff m \notin S_m \iff m \notin D$, contradiction. No complete listing exists.

6

Suppose $(0, 1]$ countable. Remove endpoint one and relist its subset $(0, 1)$ as $r_1, r_2, \ldots$. Choose canonical decimals with no eventually-all-nine tail. Removing one matters because its canonical expansion is $1.000\cdots$, not $0.999\cdots$. Now each $r_n = 0.a_{n1}a_{n2}\cdots$.

Let $b_k = 3$ when $a_{kk} \neq 3$, and seven otherwise, and put $x = 0.b_1b_2\cdots$. It lies in $(0, 1)$ with a canonical decimal using only three and seven. If $x = r_m$, their canonical expansions would differ at digit m, impossible. The missing x contradicts the list. Thus $(0, 1]$ and its real superset are uncountable.

7

Rationals are countable. Map the countable product $(\mathbb{Z} \setminus \{0\}) \times \mathbb{N}^+$ onto nonzero rationals by $(p, q) \mapsto p/q$; remove duplicates and add zero. Irrationals are uncountable, since otherwise their union with the countable rationals would make the reals countable.

For finite subsets, let $\mathcal{P}_n = \{S \subseteq \mathbb{N} : |S| = n\}$. Each has a unique increasing tuple $(a_1 < \cdots < a_n)$, injecting it into the countable finite product $\mathbb{N}^n$. These layers have explicit enumerations and their countable union includes exactly the finite subsets, so it is countable. All subsets form $\mathcal{P}(\mathbb{N})$, uncountable by the diagonal proof in solution five.

8

Triangular numbers are strictly increasing and unbounded, so exactly one s satisfies $T_s \leq z < T_{s+1}$. Set $m = z - T_s \in \{0, \ldots, s\}$, $n = s - m$. Then $\pi(m, n) = z$. Any other preimage would have its diagonal in the same unique interval, forcing the same s, m, n. This proves the general inverse, beyond a finite table.

For $z = 18$, $s = 5, T_5 = 15$, so $(m, n) = (3, 2)$. For $z = 35$, $T_7 = 28 \leq 35 < T_8 = 36$, giving $(7, 0)$. Taking diagonal eight would exceed the index.

Appendix 3 solutions (continued)

9

The set $\{0, 2, 5\}$ encodes as $1 + 4 + 32 = 37$. Since $45 = 32 + 8 + 4 + 1$, its inverse support is $\{0, 2, 3, 5\}$. Unique binary expansion proves injectivity, and every nonnegative integer's finite support proves surjectivity.

An infinite subset is unbounded. Its partial sums include arbitrarily large terms 2^i, so $\sum_{i \in S} 2^i$ cannot converge to a finite integer. This encoding concerns finite subsets only; their countable union differs from the full powerset.

10

For arbitrary $f : A \to \mathcal{P}(A)$, define $D = \{a \in A : a \notin f(a)\}$. If onto, some d has $f(d) = D$, forcing $d \in D \iff d \notin D$. For empty A, its powerset contains the empty set, and the empty function cannot be onto.

The singleton injection gives $|A| \leq |\mathcal{P}(A)|$. Equality would provide a bijection, hence the excluded surjection, so the inequality is strict. The diagonal argument works against every proposed function; merely finding one non-surjective function would not suffice.

11

With $A_0 = A \setminus g(B), A_{n+1} = g(f(A_n)), C = \bigcup_n A_n$, an element outside C lies outside A_0, so has a unique preimage under g. Thus both branches are defined. Within-branch injectivity follows from f and from the inverse of the injection g. A cross equality $f(c) = g^{-1}(a), c \in C, a \notin C$, would force $a = g(f(c)) \in C$, contradiction.

For $b \in B$, if $g(b) \notin C$, then $h(g(b)) = b$. If it belongs to C, it is not in A_0, so $g(b) = g(f(c))$ for some $c \in A_n$. Injectivity gives $b = f(c) = h(c)$. Thus h is onto.

For $A = \mathbb{N}_0, B = 2\mathbb{N}_0$, take $f(n) = 2n$ and inclusion g. Then A_0 is the odd positives and $A_k = \{2^k(2j + 1) : j \geq 0\}$. Their union is all positive integers, excluding zero. The resulting bijection fixes zero and sends each positive n to $2n$.

12

Suppose S, T first differ at k, with $k \in S$ and $k \notin T$. The encoded difference is at least

$$\frac{2}{3^{k+1}} - \sum_{i>k} \frac{2}{3^{i+1}} = \frac{2}{3^{k+1}} - \frac{1}{3^{k+1}} = \frac{1}{3^{k+1}} > 0.$$

The entire tail cannot cancel the leading difference, proving injectivity.

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github.com/xinhi6716/computer-science-self-study

Appendix 3 solutions (continued)

Fix a rational enumeration and set $C_x = \{i : q_i < x\}$. For $x < y$, choose integer N with $N(y - x) > 1$, then integer $Nx < k < Ny$. The rational k/N has an index in C_y but not C_x. Thus the reverse map is injective. CSB supplies a bijection and equal cardinalities; these two injections need not themselves be inverses.

13

A degree-d integer polynomial is specified by $d + 1$ integer coefficients, a countable finite product. The union over finite degrees is countable. Excluding the zero polynomial, a degree-d polynomial has at most d distinct roots: factor out $x - a$ at a root and apply induction to the remaining degree $d - 1$ polynomial. The countable union of finite real-root sets contains all algebraic reals and is countable.

If transcendental reals were countable too, their union with algebraic reals would make $\mathbb{R}$ countable, contradiction.

Finite programs can be listed by length and lexicographic order. Their total binary functions form at most a countable image. All binary sequences, viewed as functions on $\mathbb{N}_0$, are uncountable by diagonalization, so some lie outside that image. This global cardinality proof provides no decision procedure for computability of a specified function.

14

The values are $\phi(0) = (0, 0), \phi(2) = (1, 1), \phi(-2) = (-1, 1)$, all on the circle. The inverse $2x/(2 - y)$ gives zero, two, and minus two respectively.

For any $Q \neq P$, $2 - y > 0$. Take $t = 2x/(2 - y)$. The circle identity $x^2 = y(2 - y)$ yields $t^2 + 4 = 8/(2 - y)$. Substitution gives $4t/(t^2 + 4) = x$ and $2t^2/(t^2 + 4) = y$. This proves onto with a unique inverse.

The full circle is the punctured circle plus one new point. Apply ϕ^{-1} to obtain $\mathbb{R}$ plus one element, then send that element to zero, nonnegative integer n to $n + 1$, and other reals to themselves. The moved integer part and fixed remaining part are disjoint, and each output has one preimage. This explicitly proves the full circle equinumerous with $\mathbb{R}$.

References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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https://github.com/xinhi6716/computer-science-self-study